

Manual Muscle Testing (MMT)

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- **Introduction**

- **Most important part of assessment and examination.**
- **Helps in**
 - **Diagnosis**
 - **Prognosis**
 - **Making the proper treatment protocol**
- **This is a test to evaluate muscle strength**
- **A muscle is made to do it's main function and then strength and power is assessed**
- **Used in ortho and neuro cases**

- **Introduction**

- **There are many muscle that work on a joint simultaneously, but each muscle has its own prime function.**
- **But in some cases, the prime function is taken up by other muscle during movement, this is called **Trick Movement**.**

- **Basic components of MMT**

- **Position**
- **Fixation/Stabilization**
- **Resistance**
- **Proper knowledge of muscle work**

- **Position**

- **Position of the subject is very important**
- **Position depends upon whether we have to test for against gravity or eliminating gravity.**
- **In small muscles like hand, feet and forearm rotators gravity doesn't effect the movement that much.**

- **Stabilization**

- **Stabilization is necessary to the parts which are stationary.**
- **Usually the origin site of the muscle**
- **Mostly body weight itself is responsible for stability but sometimes mechanical support is needed.**
- **The test can be done in two ways.**

- **Stabilization**

- **Subject does the movement**
- **Examiner places the part precisely in the desired position and ask the patient to hold the position against gravity and some external force.**
- **If a multi-joint muscle has to be tested**
 - **Hams is a two joint muscle (Hip extensor & Knee flexor)**
 - **Primary action of Gluteus maximus as a hip extensor can be isolated from hams by keeping the knee in flexion while testing Gluteus maximus**

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- **Resistance**

- **Resistance is also applied during the motion, as it helps to understand the power of muscle.**
- **Resistance is always applied at the insertion of the muscle.**
 - **Slight or minimum resistance (60-70%) - Fair**
 - **Medium Resistance (80-90%) - Good**
 - **Maximum Resistance (100%) - Normal**

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- **Grading**

- **There are two grading scales used to assess the power and strength of the muscles**

- **Oxford scale**
- **Lovett scale**

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- **Oxford Scale**

- **0 - No contraction**
- **1 - Flicker contraction**
- **2 - Full /Partial ROM eliminating gravity**
- **3 - Full ROM against gravity**
- **4 - Full ROM against moderate resistance**
- **5 - Full ROM against maximum resistance**

- **Lovett Scale**

- **Gone - No contraction**
- **Trace - Muscle can be felt tighter but can't produce movement**
- **Poor - Movement without gravity**
- **Fair - Movement against gravity**
- **Good - Movement against gravity & moderate resistance**
- **Normal - Movement against gravity and severe resistance**

- **Limitation of MMT**

- **Can not be done in UMNL due to spasticity and synergy**
- **Pain**
- **Muscle Spasm**
- **Mental retardation**
- **Un-cooperative patient**

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